

Health and Livelihoods

A unit level understanding households and their health outcomes.

Introduction

The type of amenities and resources that households access can have health implications for its members, particularly in relation to exposure to pollutants and degree of hygiene the amenity allows. More specifically, the type of cooking gas fuel used, the source of drinking water, the type of latrine and drainage used are daily exposures that can have long term or short term effects on health. But the possibility of a trend doesn't end here. Changes in expenditure on health can also be an outcome of low quality household facilities.

The 71st round of NSS released social consumption data on health used variables that sought to understand the profile of morbidity, linking it to demographic characteristics of religion, caste, gender and lifestyle conditions such as drainage and latrine type, source of drinking water as well as cooking gas fuel used at the household level while also covering health expenditure.

Our paper specifically looks at health expenditure and household facilities. The tables that we have produced look at trends on an all-India level, so the variation in the actual data itself was pretty enormous.

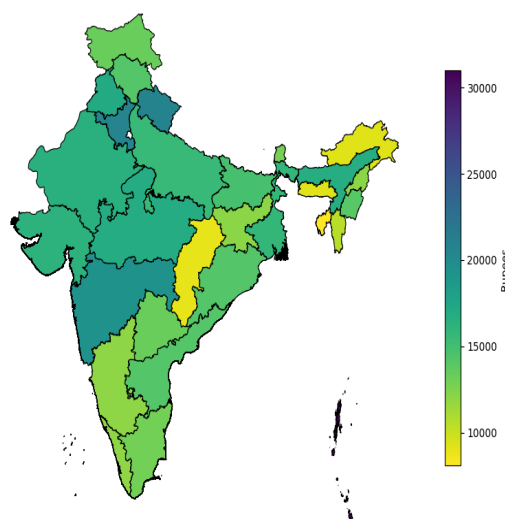
Table1:

Comparing Average Monthly Consumption and Yearly Health Expenditure Across different Social Groups with Weights								
Socio-Economic Indicators	Count	Percentage	AMC	YHE	OPE	Average Stay	YHE/AMCE	OPE/AMCE
Religion								
1 Hinduism	44462.0	77.4	9353.4	21186.1	16908.8	5.9	2.3	1.8
2 Islam	7923.0	13.8	7980.7	14675.9	13768.4	5.2	1.8	1.7
3 Christianity	3120.0	5.4	11282.5	15978.8	14131.2	5.5	1.4	1.3
4 Sikhism	1017.0	1.8	14945.2	48395.5	41908.2	7.8	3.2	2.8
5 Jainism	158.0	0.3	16355.1	41375.2	33617.0	5.4	2.5	2.1
6 Buddhism	531.0	0.9	6858.1	12880.8	12316.1	5.1	1.9	1.8
7 Zoroastrianism	11.0	0.0	12047.7	22490.4	19487.1	8.6	1.9	1.6
9 Others	234.0	0.4	8341.9	9189.8	7597.0	6.7	1.1	0.9
Social Group								
1 ST	6397.0	11.1	6935.0	12525.2	9248.5	5.3	1.8	1.3
2 SC	9838.0	17.1	7312.8	12265.8	11214.3	5.7	1.7	1.5
3 OBC	22980.0	40.0	8736.4	17015.0	15396.7	5.5	1.9	1.8
9 Others	18259.0	31.8	11442.0	30455.6	22797.0	6.2	2.7	2.0
Sector								
1 Rural	31134.0	54.2	7575.6	15696.1	13793.4	5.6	2.1	1.8
2 Urban	26322.0	45.8	13209.7	31305.7	23443.8	6.3	2.4	1.8

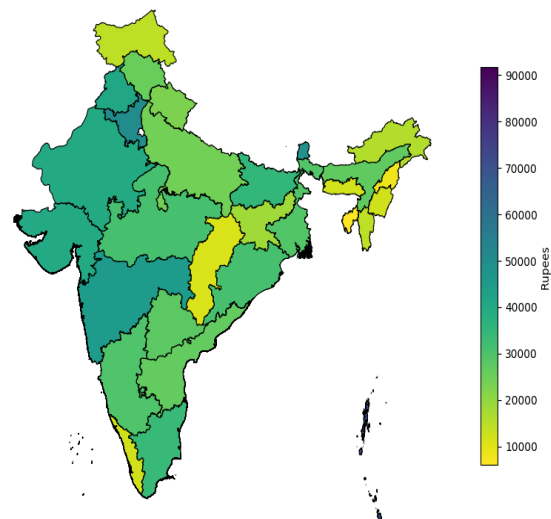
Table 1 gives us an overview of those households where there were medical expenses during the period under study. This table forms the basis from which all further analysis has been conducted. We have in the table an analysis of how these households have spent on health during this period.

We introduce here a few terms that we will be using throughout the course of our paper. AMC (alternatively AMCE) stands for annual monthly consumption expenditure. YHE stands for yearly health expenditure. OPE stands for Out of Pocket Expenditure, which is calculated by adding insurance premiums paid by households to YHE, and subtracting the amount reimbursed for medical claims for that year.

Statewise Average Monthly Consumption Expenditure



Statewise Average Yearly Health Expenditure

**Table 2:**

This table shows the characteristics of the individuals from the sample.

Distribution of individual characteristics of the sample			
Socio-Economic Indicators		Count	Percentage
Relation to the haed			
1	Self	65932	19.8
2	Spouse of Head	54819.0	16.5
3	Married child	24876.0	7.5
4	Spouse of Married Child	22801.0	6.8
5	Unmarried child	105763.0	31.8
6	Grandchild	38462.0	11.5
7	Father/mother (inc. In laws)	8877.0	2.7
8	Brother/Sister (inc. In laws)	11187.0	3.4
9	Servant/Employees	387.0	0.1
Age			
0-15	Children	86512.0	26.0
15-30	Youth	99027.0	29.7
30-45	Adult	62726.0	18.8
45-60	Elderly	43884.0	13.2
60 +	Old	23362	7.013425237
Sex			
1	Male	168697.0	50.6
2	Female	164407.0	49.4
Marrital Status			
1	Never Married	151672	45.53293
2	Currently Married	164382	49.34855
3	Widowed	15958	4.790696
4	Divorced	1092	0.327826

Education Levels			
1	Illiterate	7	0.002101
2	Literate without schooling	102994	30.91947
3	Literate without formal schooling	1229	0.368954
4	Literate through TLC	929	0.278892
5	Other	262	0.078654
6	Literate with formal schooling	1091	0.327525
7	Below Primary	46821	14.05597
8	Primary	43515	13.06349
9	Middle	45949	13.79419
10	Secondary	35445	10.64082
11	Higher Secondary	25463	7.644159
12	Diploma Secondary	1688	0.506749
13	Diploma Higher Secondary	1672	0.501945
14	Diploma Graduate	1104	0.331428
15	Graduate	18739	5.62557
16	Post Graduate and Above	6196	1.86008
Insurance Coverage			
1	Government Insurance Scheme	41027	12.316574
2	Private Employer Insurance	4230	1.269874
3	Companies	4332	1.300495
4	Others	645	0.193633
5	Not Covered	282868	84.918824

Table 3:

Impact of health facilities on yearly health expenditure, adjusting for insurance and using weights											
Health Facilites		Count	Percentage	AMC	AYHE	OPE	Days spend in hospital	Rural		Urban	
								Public	Private	Public	Private
Clean Fuels											
0	Clean	26187.00	45.60	12892.42	32619.53	24238.36	6.28	1.05	2.76	1.44	2.76
1	Unclean	31269.00	54.40	7018.94	12807.79	12018.62	5.47	1.13	2.90	0.99	2.82
Drainage											
0	Closed	17430.00	30,3	13822.80	32589.76	22906.63	6.29	0.96	2.62	0.88	2.48
1	Open	40026.00	69,7	7570.16	15833.78	14368.69	5.59	1.13	2.91	1.51	3.28
Laterine											
0	Hyginic	28761.00	50.10	12085.93	31820.33	24387.61	6.48	1.47	3.14	1.35	2.40
1	Unhygienic	28761.00	49.90	7143.78	11798.45	10917.91	5.26	1.03	2.69	1.13	3.08
Drinking water											
0	Clean	22208.00	38.70	7810.08	15101.10	14050.95	5.12	1.12	2.85	1.27	2.53
1	Dirty	35248.00	61.30	10844.58	26125.87	19551.67	6.51	0.87	3.50	1.06	1.86

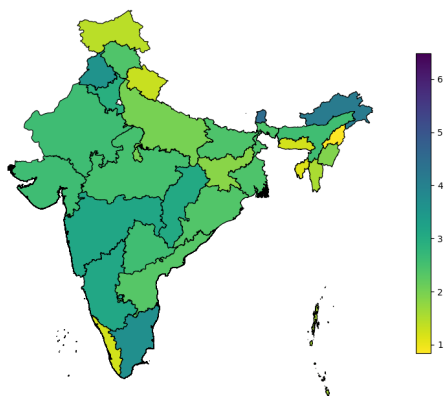
Table 3 compares the yearly health expenditure, monthly consumption expenditure and out of pocket expenditure based on the different resources people have access to.

We notice that the difference between AYHE and OPE is higher in the case of households which consume clean resources (except for drinking water. Here, dirty drinking water corresponds to the source river/canal and the category of others.) as compared to households which don't. This could indicate that there is a larger proportion of households paying insurance premiums that aren't reimbursed in the case of those who use clean resources and that there is a higher proportion of households that are being reimbursed in the case of those who use unclean resources.

Contrastingly, we notice that the difference between the AYHE and OPE is higher in the case of households which have dirty drinking water vs the ones who have clean drinking water.

This might be surprising as it seems to indicate that there is a higher proportion of people who are getting reimbursed in the case of those who have clean drinking water vs those who don't. However, we must take into account that there could be an intersection between those who have clean drinking water and say, those who have open drainage systems.

Statewise Out of Pocket Expenditure by Monthly Consumption



Comparing the States with the Highest and Lowest Clean Drinking Water

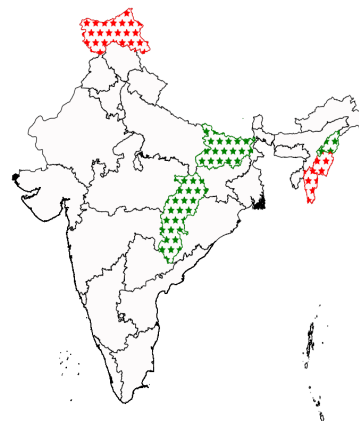


Table 4:

Impact of health facilities on yearly health expenditure on relevant diseases. Unweighted (Since sample size reduces; does not allow weighting by Stratum-- PSU size low)											
Health Facilities		Count	Percentage	AMC	AYHE	OPE	Days spend in hospital	Rural == 1		Urban == 2	
								Public == 3	Private == 5	Public	Private
Clean Fuels (5,37,38)											
0.00	Clean	702	42	12283.19	25232.31	22882.68	6.54048	0.92	1.64	0.696	2.53
1.00	Unclean	968	58	7263.603	12914.48	12804.96	6.226806	1.14	2.84	1.13	3.12
Drainage (46,47,48,1,2,3,4)											
0.00	Closed	999	34.8	13673.85	37512.16	30476.67	7.912826	1	4.14	0.987	2.43
1.00	Open	1873	65.2	8485.876	24379.26	21341.77	8.238121	1.88	3.77	1.45	2.76
Laterine (46,47,48,1,2,3,4)											
0.00	Hygienic	1516	53.8	12562.10686	35998.9041	30048.83799	8.204485488	2.61	4.59	0.904	2.41
1.00	Unhygienic	1356	47.2	7750.791298	21064.24594	17930.84161	8.036162362	1.35	3.45	1.99	3.12
Drinking water (11,12,40,42)											
0.00	Clean	5503	97.5	9031.051608	13732.45564	14971.89819	5.713299419	0.921	3.13	0.866	2.35
1.00	Dirty	142	2.52	7281.197183	8867.626761	7163.101695	5.556338028	0.742	7.03	0.29	0.955

In this table, we have the amount spent by various households with differing resources on their health when afflicted with related diseases (the numbers in the brackets represent the codes in the schedule for the diseases we have looked at). We elaborate on what kinds of diseases below. While looking at sources of fuel, we decided to see if there was a relation between using unclean sources of fuel and higher incidences of respiratory illnesses, particularly chronic illnesses such as bronchial asthma, or cough. While looking at access to drainage and latrines, we tried to understand if access to closed drainage or a higher standard of latrines affected incidences of genito-urinary illnesses and other related diseases. While looking at access to clean sources of drinking water, we tried to understand if drinking unclean water was related to an increase in spending on treatment of gastrointestinal diseases.

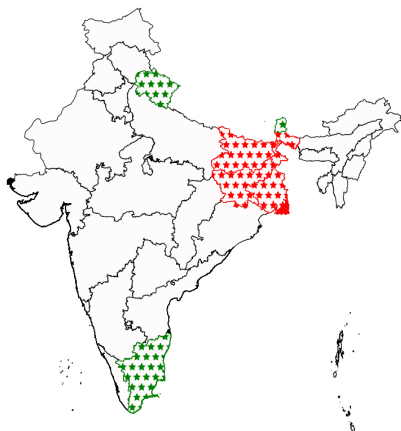
Generally speaking, we see that people that avail private medical care end up spending significantly more than people that avail public health facilities. We also see that people that lack facilities such as clean drinking water, or hygienic latrines have a smaller monthly consumption expenditure than people that have access to these resources. This points to the fact that inequality in access to resources might be tied to inequality in other domains (such as income) as well.

A number that jumps out immediately when we look at the table is the 7.03 under the OPE/ACM column for people in rural areas that went to private hospitals and had dirty water to drink. Now initially it appears as if this seals the case for dirty drinking water impacting health adversely, until we see that the size of the sample here is pretty small (all households that have access to unclean drinking water and had illnesses during the record period total up to 142. Subsetting further will only reduce the size), so it may be swayed very easily by any one large observation.

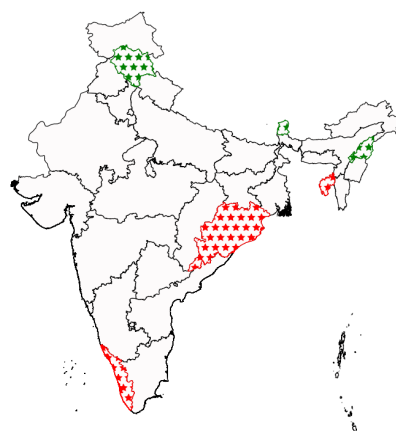
Differences in quality across the 4 facilities

The indoor air quality guidelines of the WHO have classified LPG and electricity as cleaner sources of energy while traditional cooking fuels like gobar gas, charcoal, coal, coke, firewood and chips and dung cake are causes of indoor air pollution. So our classification of clean and unclean are made accordingly.

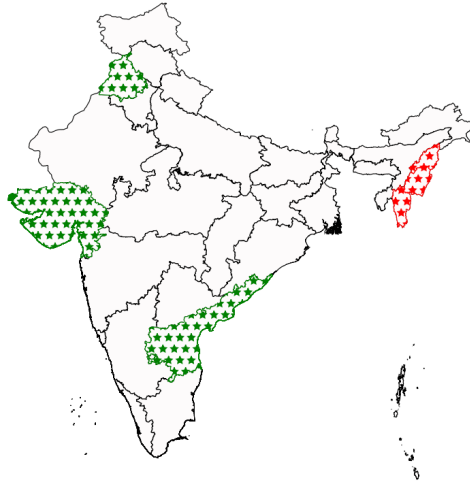
Comparing the States with the Highest and Lowest Green Cooking Fuels



Comparing the States with the Highest and Lowest Latrine Facility



Comparing the States with the Highest and Lowest Drainage



In terms of latrine facilities, this is our classification:

Clean -septic tank/flush and others

Unclean - pit, service and no latrine

Drainage facilities:

Clean- closed pucca, underground

Unclean- open kutchha, open pucca and no drainage

What We May Draw From This Data:

The data we had was extremely detailed, and also had a lot of variability in it. We wanted to understand the relationship between access to resources and spend on health for a given household. These relationships are quite complicated, especially given that one household may have, for instance, access to LPG gas as a fuel, but may not have a covered drainage system. We have tried putting together a simple linear regression analysis to understand the effect of household facilities on health expenditure. Please do refer to this for further details.

Roles of Group Members:

Merging the data, adding weights, and obtaining graphs - Vaibhav

Running Regression Analysis - Neha

Descriptive Analysis -
Neha, Vaibhav, Saranyan

Writing Report and Analysis of Table Results - Vijayshri, Vighnesh